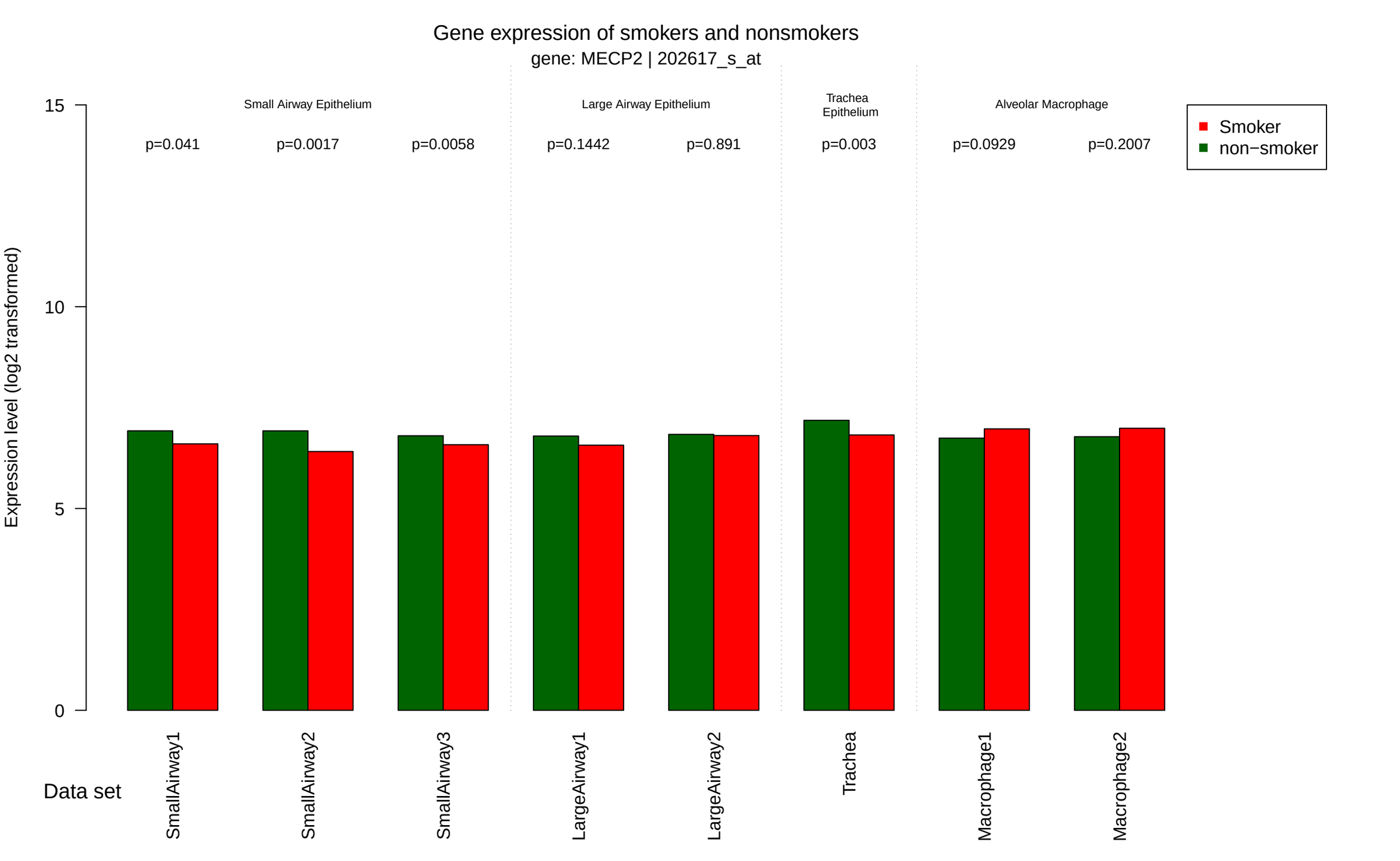
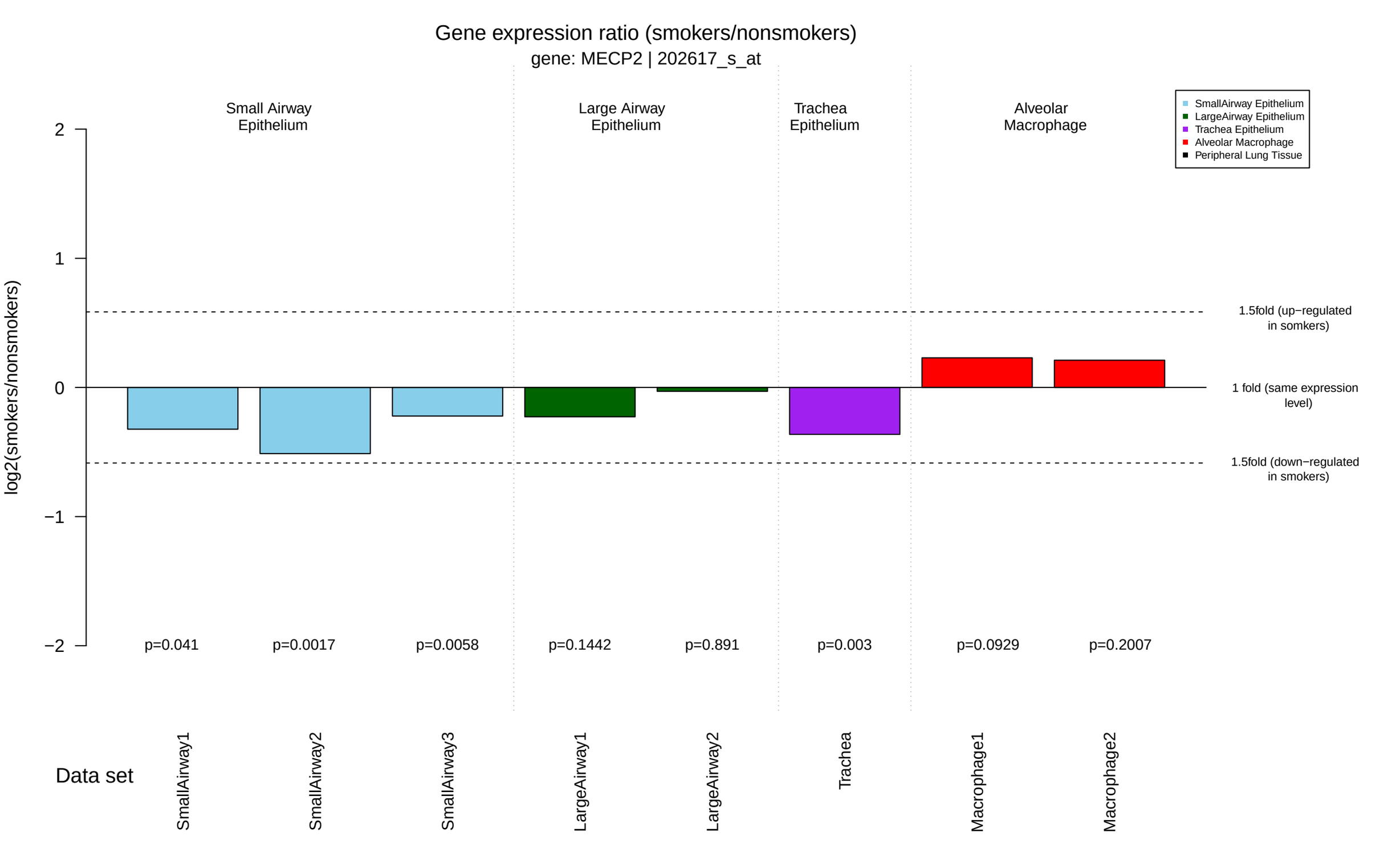
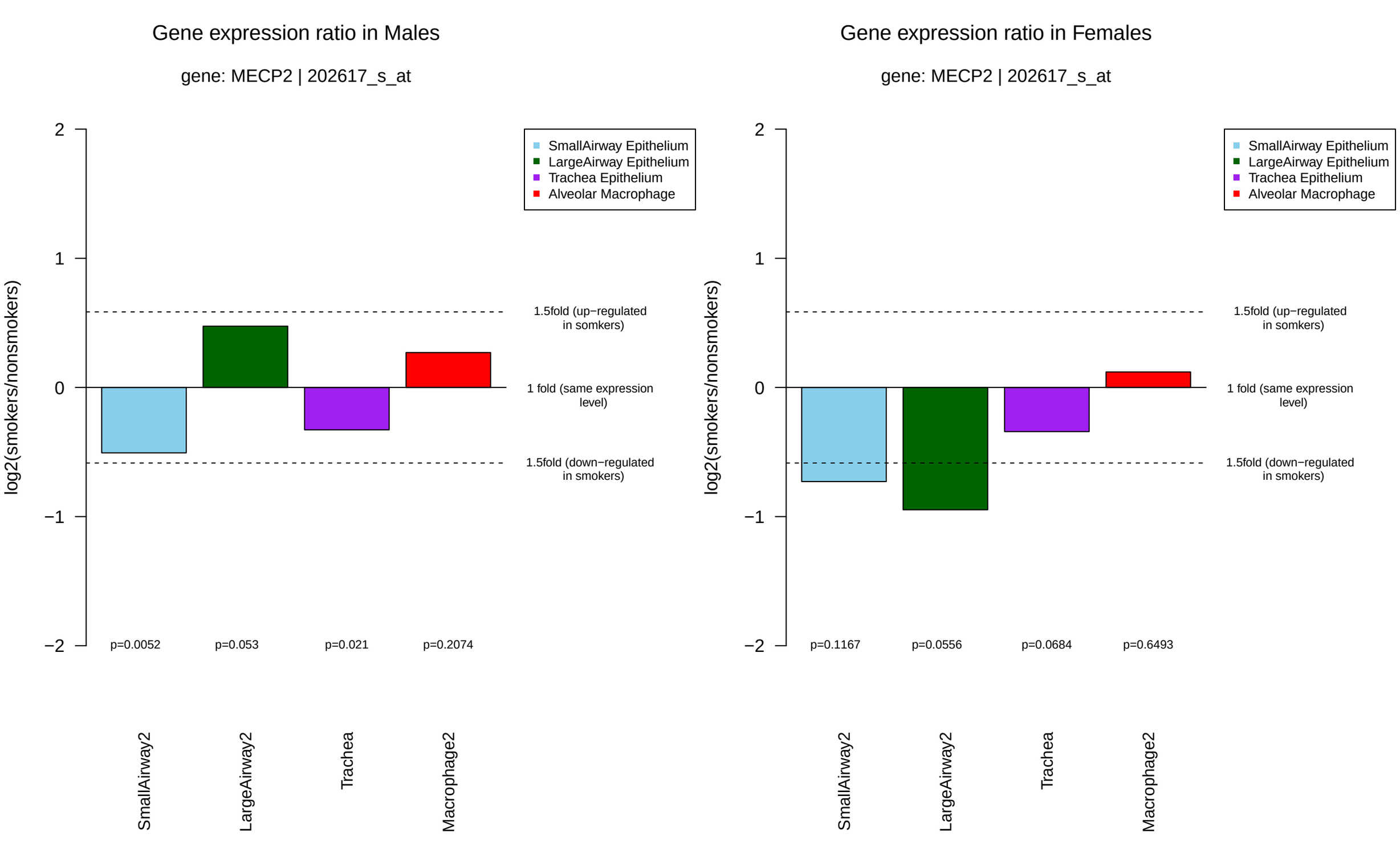
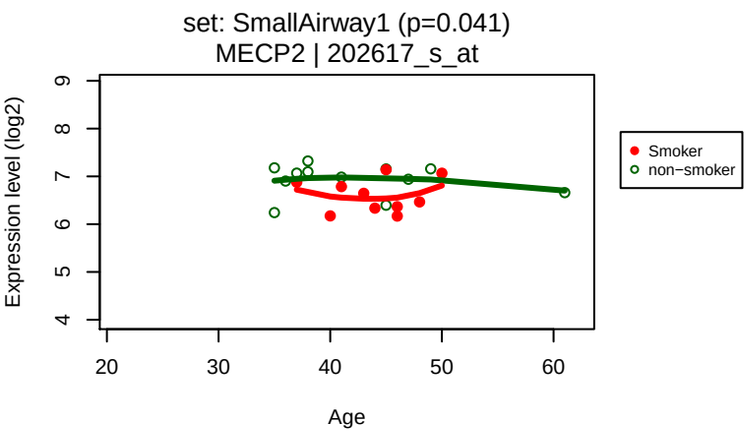
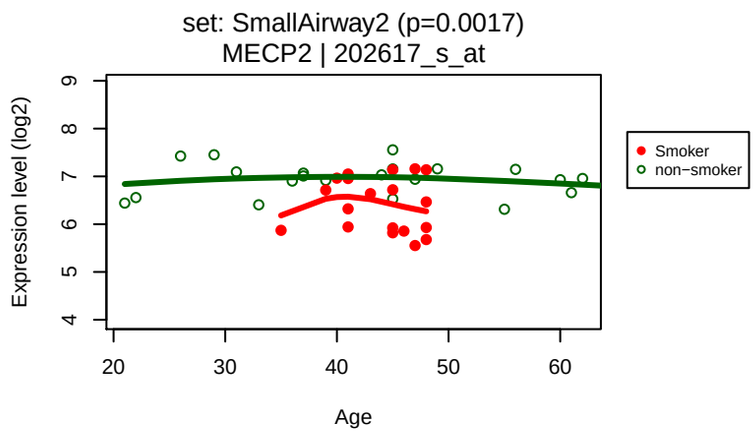
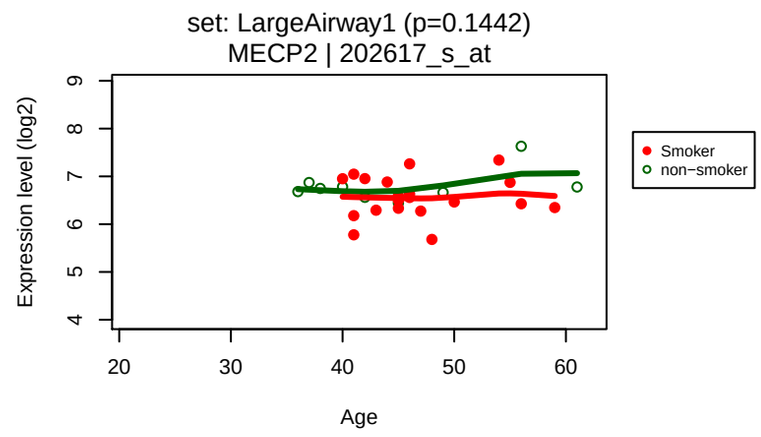

Gene expression of smokers and nonsmokers
gene: MECP2 | 202617_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.041
p=0.0017
p=0.0058
p=0.1442
p=0.891
p=0.003
p=0.0929
p=0.2007
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

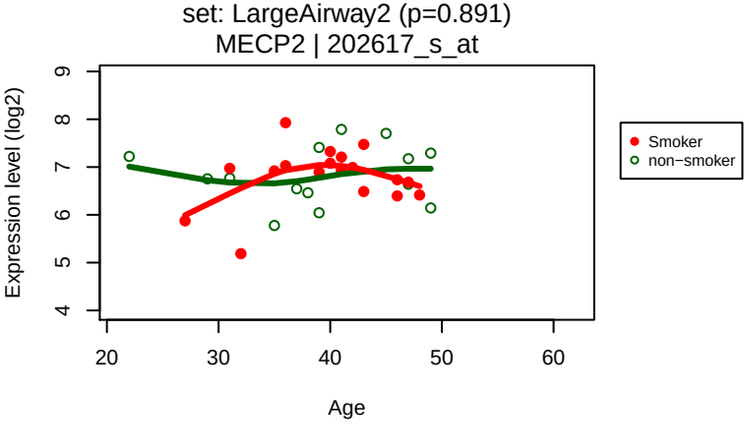

Gene expression ratio (smokers/nonsmokers)
gene: MECP2 | 202617_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.041
p=0.0017
p=0.0058
p=0.1442
p=0.891
p=0.003
p=0.0929
p=0.2007
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

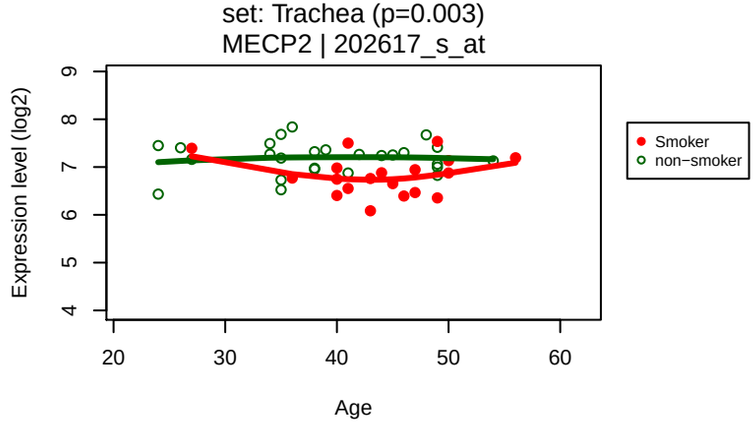

Gene expression ratio in Males
gene: MECP2 | 202617_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0052
p=0.053
p=0.021
p=0.2074
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MECP2 | 202617_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1167
p=0.0556
p=0.0684
p=0.6493
SmallAirway2
LargeAirway2
Trachea
Macrophage2

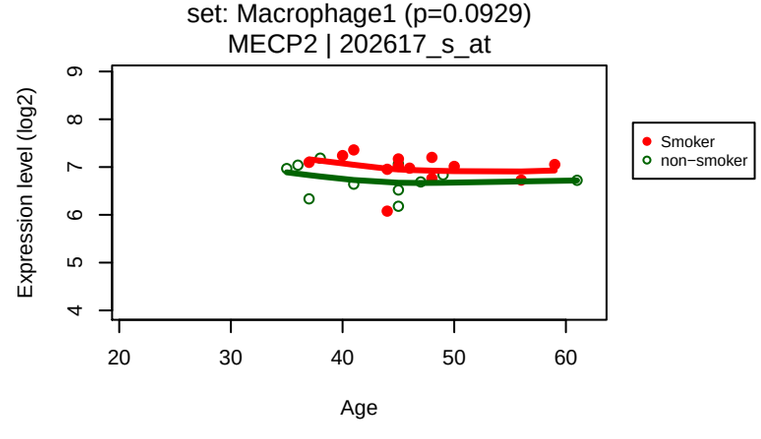

set: SmallAirway1 (p=0.041)
MECP2 | 202617_s_at
Expression level (log2)
Age
Smoker
non−smoker

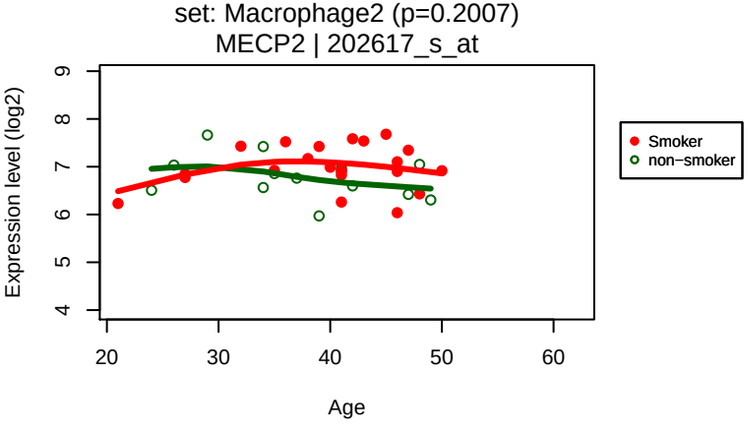

set: SmallAirway2 (p=0.0017)
MECP2 | 202617_s_at
Expression level (log2)
Age
Smoker
non−smoker

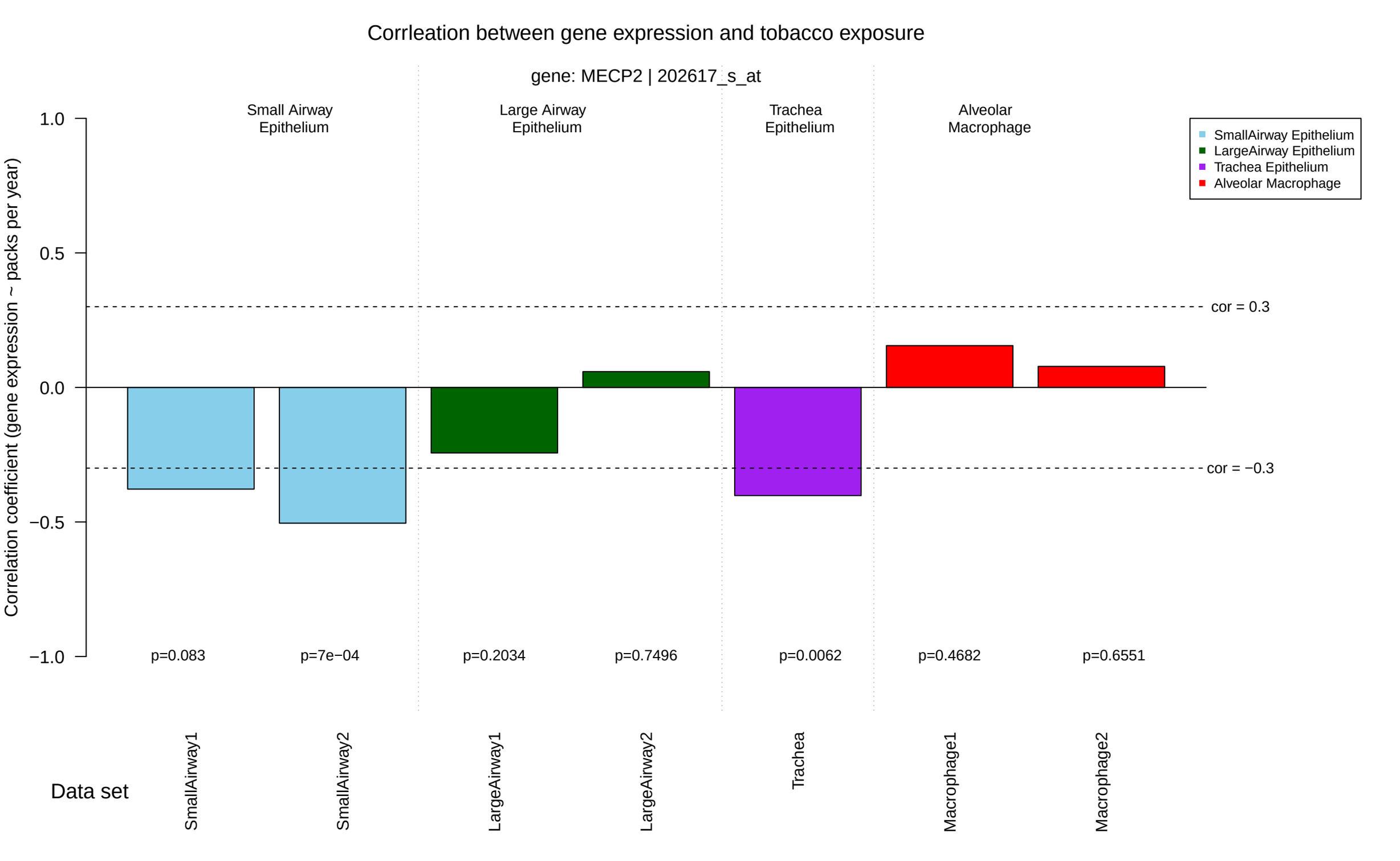

set: LargeAirway1 (p=0.1442)
MECP2 | 202617_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.891)
MECP2 | 202617_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.003)
MECP2 | 202617_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0929)
MECP2 | 202617_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2007)
MECP2 | 202617_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MECP2 | 202617_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.083
p=7e−04
p=0.2034
p=0.7496
p=0.0062
p=0.4682
p=0.6551
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2